

Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 1

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

Team Members

Aditi Kapadiya	AU2440139
Disha Gadhiya	AU2440217
Khushi Panchal	AU2440002
Harshita Dhanwani	AU2440083

Week 1

1. Introduction

The quality checking of fabric is an important job in the textile industry. Fabric can be of poor quality and fit for use due to various defects like broken yarns, knots, creases, contamination and cracks. Manual inspection can be time consuming and can sometimes be inconsistent.

The project is focused on exploring the classical image processing and texture analysis methods for the detection and localization of defects in fabrics. The AITEX Fabric Image Database (AFID) is the main database as it includes various types of fabrics, defects and ground-truth masks.

The primary objective of the project is to explore the utilization of multi-scale and adaptive texture analysis in comparison to a fixed scale analysis approach to achieve improved defect detection.

2. Literature Review

The defects can be considered as a problem of texture analysis because they break the regular texture or weave pattern of the fabric. In this problem several classical image processing techniques can be applied.

- **LBP:** Local Binary Pattern (LBP) is a local texture characterized through a comparison of the pixel with its surrounding pixels. Simple and easy to calculate, it can be used to detect the irregularities in local texture.
- **GLCM:** Gray-Level Co-occurrence Matrix (GLCM) is a way of quantifying the spatial relationship between neighbouring pixels. Fabric texture can be described by features like contrast, energy, homogeneity and correlation.
- **Gabor Filters:** Gabor filters are used for texture analysis for various orientations and frequencies. They are especially valuable for fabrics as fabric patterns tend to be periodic and directional.
- **FFT:** The image is shown in the frequency domain using Fast Fourier Transform (FFT). It can show the prevalent periodicity of a cloth texture, which can be helpful in determining appropriate analysis scales.
- **Multi-Scale and Adaptive Analysis:** There are different texture patterns for fabrics and defect sizes can vary. Use of more scales can thus enhance texture representation. An adaptive approach can also use the suitable scale parameters for each fabric and not the same parameters for all fabrics.
- **Summary:** It can be seen from the literature that LBP, GLCM and Gabor filters are effective classical texture descriptors, while FFT can give the information of texture periodicity. These techniques drive the project's comparison of fixed scale, fixed multi-scale and adaptive multi-scale texture analysis for fabric defect detection.

3. Observations

Dataset Observations:

The AITEX Fabric Image Database (AFID) was chosen for this project due to the availability of grayscale images of various fabric types, both with and without defects. The dataset also contains ground-truth masks for images that have defects, allowing for the evaluation of defect localization.

There are 7 types of fabric and 12 types of defects in the data set. The images are long-strip images, that is, images of the fabric taken in the direction of the fabric production. This configuration places a special emphasis on the importance of texture analysis in the definition of normal fabric patterns and in the detection of defects.

Image Characteristics:

Based on the exploratory analysis, the images of the fabric appear to be very textured and repeated. Fabrics vary in type, with patterns and intensities of texture. There are significant differences in defects in terms of:

- Shape
- Size
- Location
- Intensity
- Texture disturbance

Thus, it is difficult to apply a single texture-analysis scale to all categories of defects and structures of fabrics.

4. Conclusion:

In week 1 we have provided an introduction to the AITEX dataset, its fabric and defect features and classical texture analysis for fabric defect detection. The EDA also pointed to variations in texture of the fabric and the sizes of defects, which inspired a study of multi-scale and adaptive texture-analysis methods for the later phases of the project.

5. Week 2 — Preprocessing Pipeline

In Week 2, we will work on preparing AITEX AFID data for implementing texture-based fabric defect detection methods.

Tasks Planned

- Prepare the dataset: Compose fabric images and their ground-truth defect masks in a way that allows them to be used consistently throughout the project.
- Data loading and visualization: There is coding involved for loading the images and masks, and also visualizing both images and masks together to see if the regions of the defect are identified.

- Illumination normalization: Pre-processing to overcome differences in illumination of the fabric images.
- Try CLAHE (Contrast Limited Adaptive Histogram Equalization) to create better local contrast and to better distinguish the texture patterns in the fabric.
- Denoising: Use appropriate denoising methods to suppress the unwanted image noise while retaining the important image texture and defect information.
- The AITEX images are long fabric strips and have to be broken down into smaller overlapping patches, for example 256×256 pixels, to enable easier texture analysis.
- For each image patch, generate a corresponding patch from ground-truth masks with the same coordinates as the image patches to ensure the images and masks are correctly aligned.
- Preprocessing comparison: visualize and compare original images with processed images, select best image preprocessing steps for further texture analysis.
- Documentation: Commit the preprocessing code and visualisations, observations, and Week 2 progress to GitHub.

By the end of Week 2, we will have a working data-loading and preprocessing pipeline that takes the raw AITEX images and returns image patches and their associated ground-truth masks in a standard format.